Control 1 (COMP0208) Assessment 2

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Our team's specific parameters (based on Adeel's student number, 25057665):

```
--- Pitch Control System Parameters ---
Moment of Inertia (J): 0.0156 kgm^2
Max Control Voltage (u_max): 10326 V
-------------------------------------
My (Adeel) specific parameters:
J = 0.015608 kg*m^2
umax = 10326.0000 V
```

Figure 1: System parameters generated by pitch_robot.p

1 Assignment 4.1: Mathematical Modeling

1.1 Simulink Model

The model implements the five subsystem blocks and the internal back-EMF feedback loop:

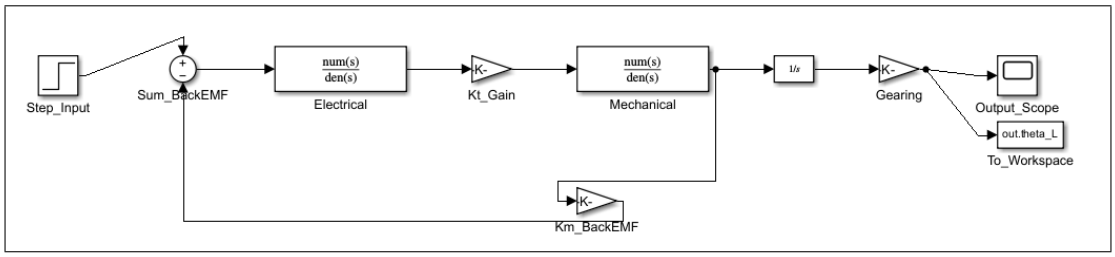

Figure 2: Simulink model representing the wind turbine pitch control system

1.2 Derivation of the Open-Loop Transfer Function $G(s)$

The open-loop transfer function $G(s) = \Theta_L(s)/U(s)$ is derived by initially translating the fundamental physical equations into the Laplace domain, and then removing some variables.

1.2.1 Fundamental Equations

The electrical armature circuit is described by Kirchhoff's voltage law:

$$u(t) - K_m\dot{\theta}_m(t) = L_m\frac{dI_a}{dt} + R_mI_a(t) \tag{1}$$

where $K_m\dot{\theta}_m$ represents the back-EMF voltage generated by the spinning motor.

The mechanical rotating shaft follows Newton's second law for rotation:

$$J\frac{d\dot{\theta}_m}{dt} = K_\tau I_a(t) - b\,\dot{\theta}_m(t) \tag{2}$$

The motor coupling relationships are:

$$T(t) = K_\tau\, I_a(t), \qquad v_{\text{back}}(t) = K_m\,\dot{\theta}_m(t) \tag{3}$$

The kinematic relationships for position and gearing are:

$$\theta_m(t) = \int \dot{\theta}_m(t)\,dt, \qquad \theta_L(t) = n\,\theta_m(t) \tag{4}$$

1.2.2 Laplace Domain Transformation

Applying the Laplace transform to the equations under zero initial conditions:

$$(sL_m + R_m)\,I_a(s) = U(s) - K_m\,s\,\Theta_m(s) \tag{5}$$
$$(Js + b)\,s\,\Theta_m(s) = K_\tau\,I_a(s) \tag{6}$$
$$\Theta_L(s) = n\,\Theta_m(s) \tag{7}$$

1.2.3 Algebraic Reduction

From Equation (6), the armature current can be expressed as:

$$I_a(s) = \frac{(Js + b)\,s\,\Theta_m(s)}{K_\tau} \tag{8}$$

Substituting Equation (8) into Equation (5):

$$(sL_m + R_m)\cdot\frac{(Js + b)\,s\,\Theta_m(s)}{K_\tau} = U(s) - K_m\,s\,\Theta_m(s) \tag{9}$$

Multiplying both sides by K_τ and collecting $\Theta_m(s)$ terms on the left-hand side:

$$[(sL_m + R_m)(Js + b)\,s + K_mK_\tau\,s]\,\Theta_m(s) = K_\tau\,U(s) \tag{10}$$

Factoring out the s from the bracket and substituting $\Theta_L(s) = n\,\Theta_m(s)$, then rearranging:

$$\boxed{G(s) = \frac{\Theta_L(s)}{U(s)} = \frac{n\,K_\tau}{s\,[(sL_m + R_m)(Js + b) + K_mK_\tau]}} \tag{11}$$

Fully expanding the denominator gives:

$$G(s) = \frac{nK_\tau}{L_mJ\,s^3 + (L_mb + R_mJ)\,s^2 + (R_mb + K_mK_\tau)\,s} \tag{12}$$

The denominator contains a free factor of s, which corresponds to the integrator block $(1/s)$ from the Simulink model which converts motor angular velocity $\dot{\theta}_m$ to angular position θ_m.

Substituting the known parameter values yields the transfer function:

```
QUESTION 4.1: Mathematical Modeling
Open-loop Transfer Function G(s):
G(s) = 0.0100 / (0.0008*s^3 + 0.0674*s^2 + 0.6500*s)
```

Figure 3: Derived open-loop transfer function G(s)

1.3 Open-Loop Stability Analysis

The stability of the system is determined by analysing the pole locations of $G(s)$. A system is asymptotically stable if, and only if, all poles have strictly negative real parts...

```
Poles of G(s):
  p1 = 0.000000 + 0.000000j
  p2 = -75.353566 + 0.000000j
  p3 = -11.053537 + 0.000000j
```

Figure 4: Locations of poles of the open-loop transfer function $G(s)$

The denominator of Equation (12) can be factored as $s \cdot (L_m J\, s^2 + \ldots)$. This means that there is a pole at $s = 0$ and two additional poles determined by the quadratic factor. The two non-zero poles lie in the open left half-plane, and so are classified as negative real parts, however, the pole at $s = 0$ lies exactly on the imaginary axis.

Since the condition of asymptotic stability necessitates strictly negative real parts for all poles ($\mathrm{Re}(p_i) < 0,\ \forall i$), and the pole at $s = 0$ has $\mathrm{Re}(p) = 0$, the open-loop system is, therefore, not asymptotically stable. Technically, it is classified as marginally stable.

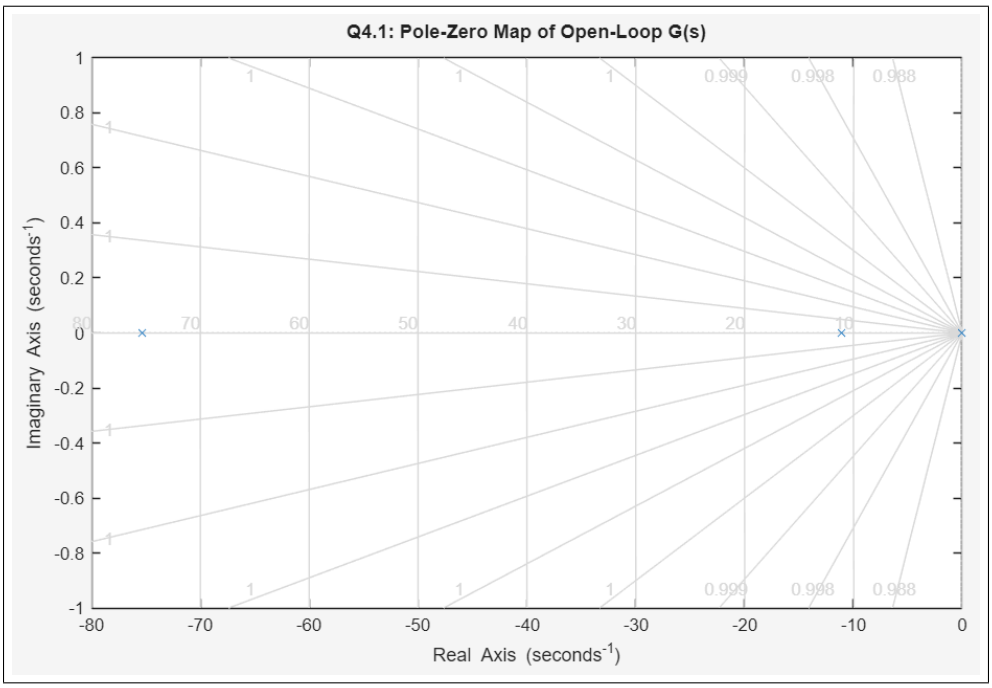

Figure 5: Pole-zero map of the open-loop transfer function $G(s)$. Light blue crosses represent poles. The pole at the origin confirms marginal stability

2 Assignment 4.2: Proportional Control & Stability

2.1 Control functions

Two transfer functions are now used:

- **Open-loop:** $G_o(s) = K \cdot G(s)$, used for frequency-domain stability analysis
- **Closed-loop:** $G_c(s) = \dfrac{K\,G(s)}{1 + K\,G(s)}$, describing the actual input–output behaviour of the control system.

2.2 P-Controller Design

2.2.1 Design Methodology

In order to find the optimal K, a numerical search was conducted over the range $[0.1,\ 5000]$, producing 50,000 candidates. Each candidates' stability was verified by checking that all poles had negative real parts.

2.2.2 Results

The optimal gain was thus identified as the largest K for which the overshoot remained strictly below 5%:

```
QUESTION 4.2: Proportional Control & Stability
Searching for optimal K where overshoot < 5% with minimum rise time...
Optimal Proportional Gain: K = 326.1000
Resulting Overshoot:       4.9982%
Resulting Rise Time:       0.2844 s
```

Figure 6: Closed-loop step response characteristics with the designed P-controller

The step response is shown in Figure 6, confirming that the overshoot constraint is satisfied while achieving the shortest feasible rise time.

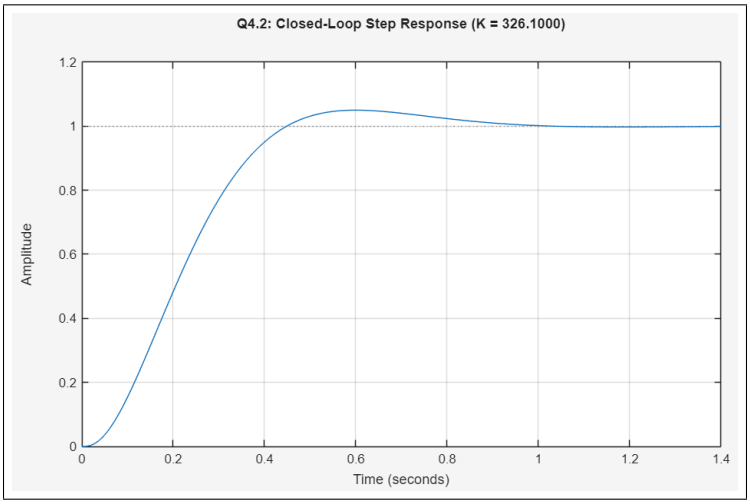

Figure 7: Closed-loop unit step response with designed proportional controller $F(s) = K$

2.3 Frequency-Domain Analysis

2.3.1 Bode Plot

The Bode plot of the open-loop transfer function $G_o(s) = K \cdot G(s)$ computes the stability margins. The resulting plot is presented in Figure 8.

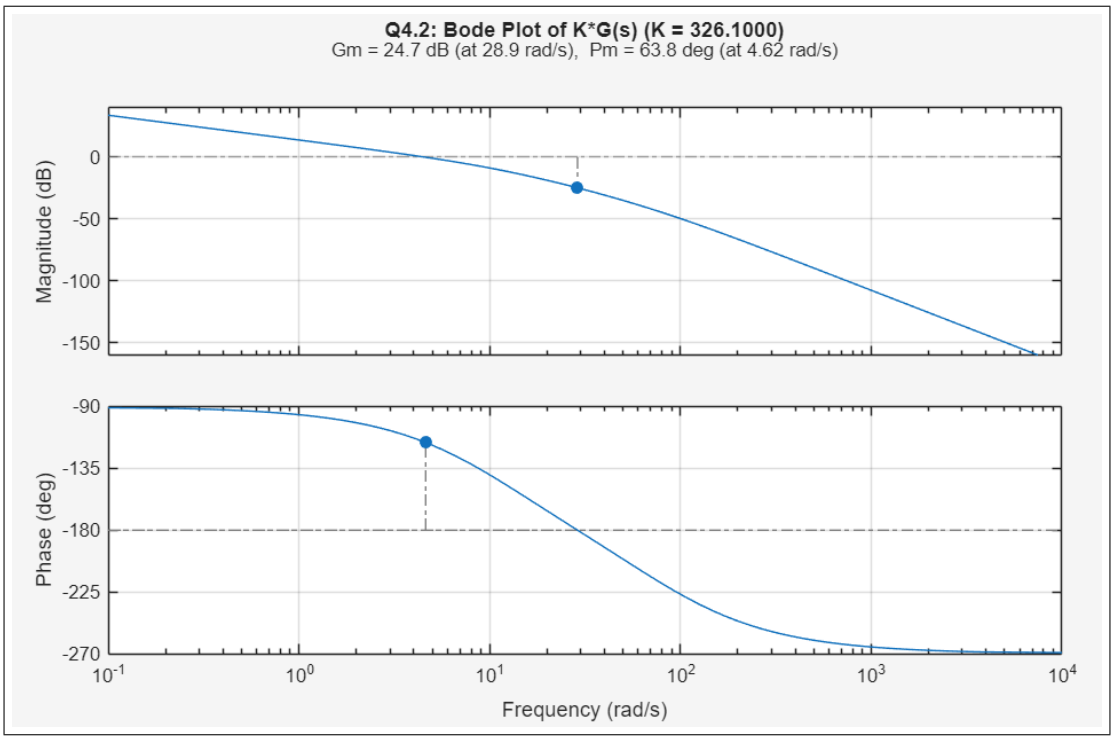

Figure 8: Bode diagram of the open-loop system $G_o(s) = K \cdot G(s)$ with the designed P-controller gain

The frequency-domain characteristics have been analysed from the Bode plot:

```
Gain Crossover Frequency:   4.6202 rad/s
Phase Margin:                      63.8074 degrees
Closed-Loop Bandwidth:             7.5463 rad/s
```

Figure 9: Cross-over frequency and phase-margin for the open loop system (including the proportional controller) and the bandwidth of the closed loop system

The **gain crossover frequency** ω_c [Figure: 9] is the frequency at which $|G_o(j\omega)| = 1$ (0 dB). Only below this frequency can the feedback loop can effectively suppress errors.

The **phase margin** φ_m represents the additional phase lag the system can tolerate at ω_c before becoming unstable [$\varphi_m = \angle G_o(j\omega_c) - (-180°)$]. The positive phase margin [Figure: 9] confirms that the closed-loop is stable.

The **closed-loop bandwidth** ω_B represents the frequency at which the magnitude drops 3 dB below its peak. This correlates to the speed of the controlled system, a higher bandwidth corresponds to faster transient response.

2.3.2 Nyquist Diagram

The Nyquist diagram of $G_o(s)$ plot maps the frequency response $G_o(j\omega)$ onto the complex plane as ω varies from 0 to ∞.

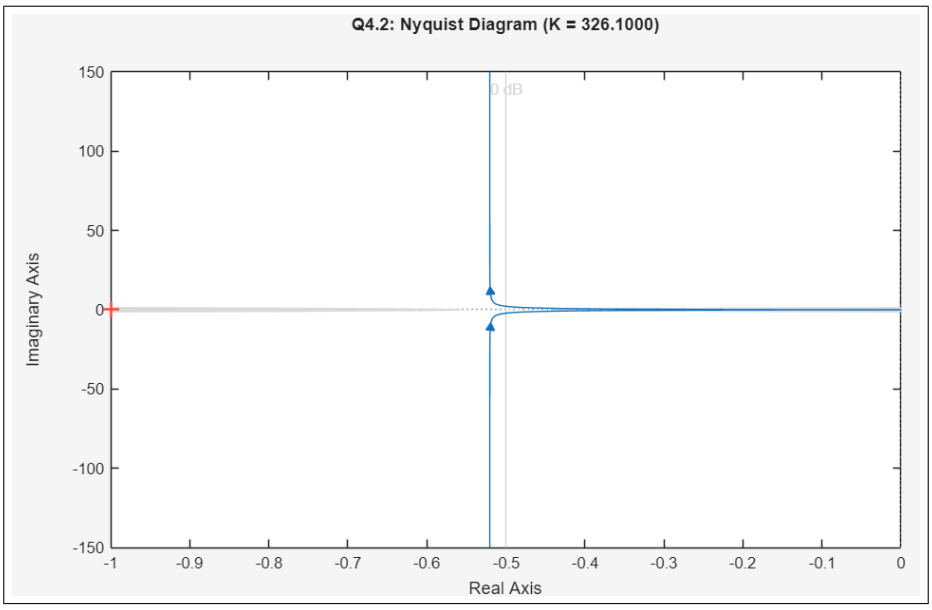

Figure 10: Nyquist diagram of $G_o(s) = K \cdot G(s)$. The critical point $(-1, 0)$ is not encircled.

As confirmed in Figure 10, the contour does not encircle the critical point, (-1, 0), at the designed gain K, again, confirming that the closed-loop is stable.

2.4 Critical Gain for Instability

2.4.1 Determination using Gain Margin

The gain margin taken from the Bode plot in Figure 8, represents the factor by which the current gain K_{opt} can be multiplied before the closed-loop system becomes unstable. The gain margin A_m is defined at the phase crossover frequency ω_p. This is the frequency where the open-loop phase reaches $-180°$.

The critical gain at which instability occurs is therefore:

$$\boxed{K_{\text{crit}} = K_{\text{opt}} \times A_m} \tag{13}$$

For all $K < K_{\text{crit}}$, the closed-loop system is asymptotically stable. When $K \geq K_{\text{crit}}$ it means there is at least one pole that crosses into the right half-plane, which represents in instability.

```
The system becomes unstable when K >= 5616.4617

Verification:
 Gain margin of G(s) alone = 17.2231 (24.7222 dB)
K_opt * Gain Margin = 326.1000 * 17.2231 = 5616.4617
```

Figure 11: Critical gain calculations using gain margin

2.4.2 Verification using Root Locus

The root locus of $G(s)$, plotted in Figure 12, traces the trajectories of the closed-loop poles as K varies from 0 to ∞. The three poles originate at the open-loop pole locations.

As K increases, two of the plots break and eventually cross the imaginary axis into the right half-plane.

The red markers in Figure 12 indicate the closed-loop pole positions at $K = K_{\text{crit}}$. Two poles lie exactly on the imaginary axis ($\text{Re}(p) = 0$), confirming our previously derived critical gain.

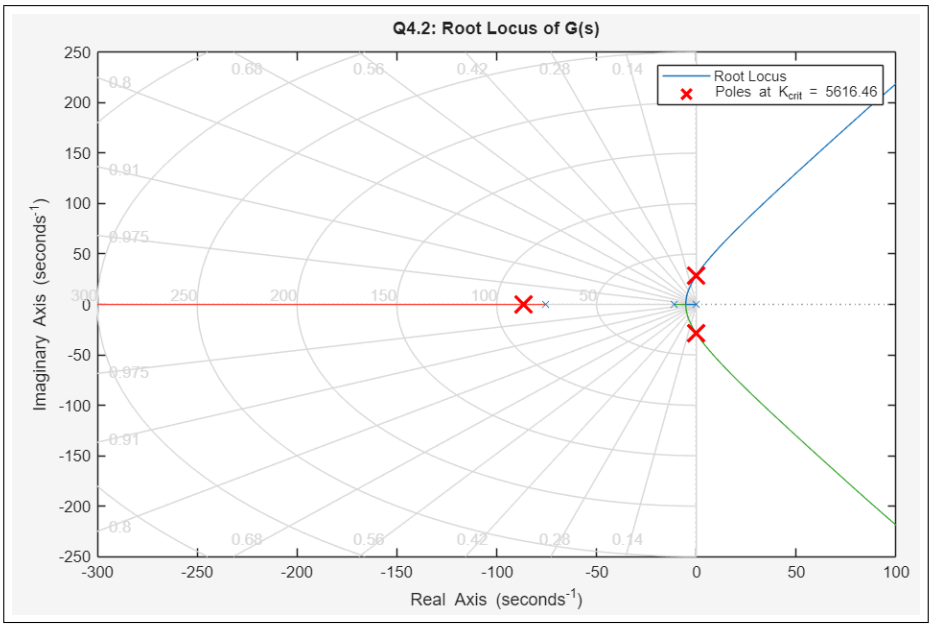

Figure 12: Root locus of $G(s)$. Red markers indicate the closed-loop poles at $K = K_{\text{crit}}$, located on the imaginary axis.

3 Assignment 4.3: Lead-Lag Compensator Design

3.1 Design Motivation and Parameter Selection

3.1.1 Target Specifications

The objective is to increase the crossover frequency to $\omega_c \approx 18.48$ rad/s, which is 4 times the bandwidth of the P-controller in Assignment 2. Furthermore, the system must keep the overshoot under 5% (requiring a Phase Margin of $\varphi_m \approx 65°$) and achieve a steady-state error $e_{ss} < 0.05$ for a ramp input, which requires the velocity error constant to be $K_v \geq 20$. The actuator saturation limit is $u_{max} = 10326$ V.

3.1.2 Lead Compensator Parameters ($\beta = 0.0723, \tau_D = 0.2012$)

To achieve the 4x speedup, the uncompensated system suffers a severe phase drop at the new target crossover frequency.

- $\beta = 0.0723$ was chosen to provide the maximum phase lead required to restore the Phase Margin to over 65° (actual: 74°).
- $\tau_D = 0.2012$ was calculated to place this phase peak exactly at the target crossover frequency of 18.48 rad/s for maximum effectiveness.

3.1.3 Proportional Gain ($K_p = 647.99$)

This gain was explicitly calculated to ensure that the magnitude of the open-loop system, combined with the Lead compensator, crosses exactly at 0 dB (a magnitude of 1) at the target crossover frequency ($\omega_c = 18.48$ rad/s).

3.1.4 Lag Compensator Parameters ($\gamma = 0.4985, \tau_I = 0.5411$)

Following the lecture definition $F_{lag}(s) = \frac{\tau_I s+1}{\tau_I s+\gamma}$ (where $0 < \gamma < 1$), the lag compensator provides a low-frequency gain boost of $1/\gamma$.

- $\gamma = 0.4985$ was chosen to provide the exact additional low-frequency gain required to satisfy the steady-state error requirement ($K_v = 20$), without shifting the high-frequency crossover point.
- $\tau_I = 0.5411$ was chosen to place the corner frequency one decade below the crossover frequency ($\omega_c/10$) so that the phase lag does not significantly degrade the newly established Phase Margin.

3.2 Verification and Results

The final design resulted in an overshoot of 2.12% and a crossover frequency of 18.55 rad/s, successfully meeting the transient and bandwidth requirements. The maximum initial control signal was verified to be 8960.22 V, which safely satisfies the actuator constraint of $u_{max} = 10326$ V.

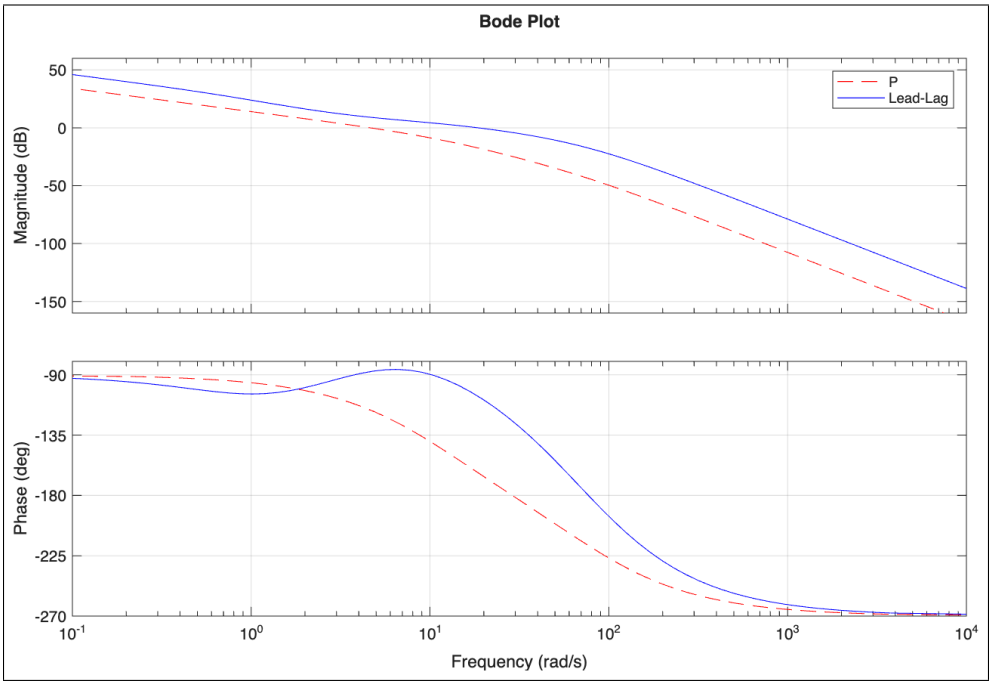

Figure 13: Open-loop Bode plot comparison between the Q4.2 P-controller and the Q4.3 Lead-Lag compensator.

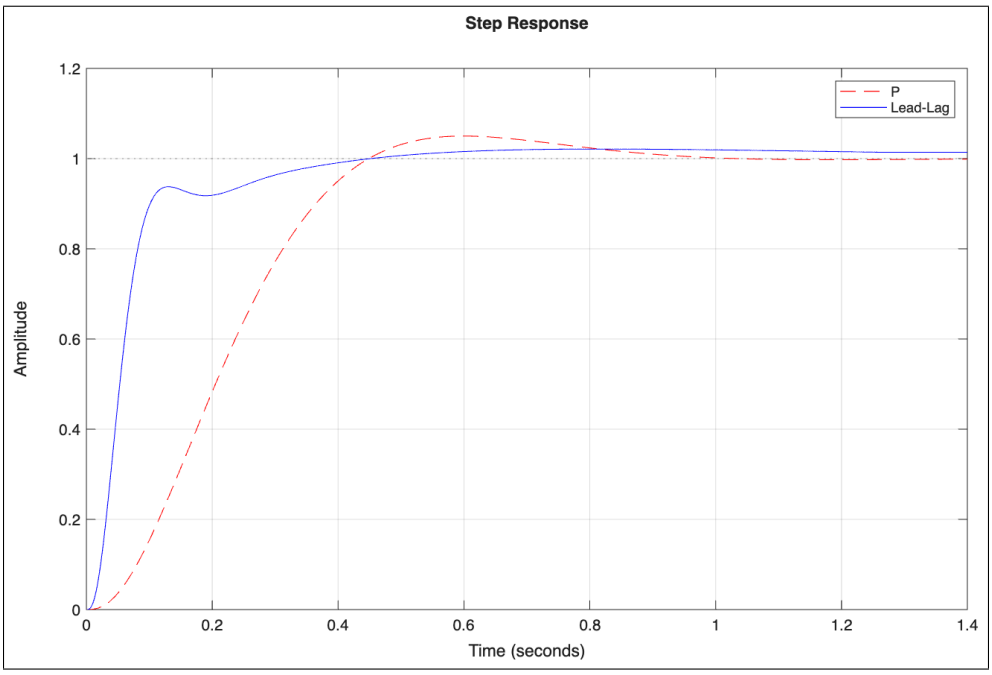

Figure 14: Closed-loop step response comparison confirming the $< 5\%$ overshoot and faster rise time.

4 Assignment 4.4: Robustness

The true transfer function is:

$$G^0(s) = G(s)\big(1 + \Delta_G(s)\big) \tag{14}$$

where $\Delta_G(s)$ is the relative model error, given by:

$$\Delta_G(s) = \frac{s+5}{20} \tag{15}$$

4.1 Robustness Criterion

If following are passed, it confirms that there are sufficient conditions for the closed-loop to remain stable despite the model error:

i) The virtual closed-loop system $G_c(s) = \dfrac{C(s)\,G(s)}{1 + C(s)\,G(s)}$ is asymptotically stable.

ii) For all frequencies ω:

$$\big|G_c(j\omega)\big| < \frac{1}{\big|\Delta_G(j\omega)\big|} \tag{16}$$

4.2 Condition i) : Stability of $G_c(s)$

The closed-loop transfer function $G_c(s)$ was made using the lead-lag compensator controller $C(s)$ from Assignment 4.3. As done previously, the pole-zero test was ran:

```
  QUESTION 4.4: Robustness Analysis
Assume the relative model error is given by: Delta_G(s) = (s + 5) / 20

Condition i): Asymptotic stability of Gc(s)
Poles of closed-loop Gc(s) with Assignment 3 controller:
  p1 = -105.998954 + 0.000000j
  p2 = -22.273686 + 29.325021j
  p3 = -22.273686 + -29.325021j
  p4 = -3.234000 + 0.000000j
  p5 = -2.268508 + 0.000000j
All poles in LHP, therefore Gc(s) is asymptotically stable
```

Figure 15: five poles of $G_c(s)$ and the stability conclusion

The closed-loop system has five poles, all with strictly negative real parts. Since $\mathrm{Re}(p_i) < 0$ for all i, the closed-loop system is almost confirmed to be asymptotically stable, as the first condition has been passed.

4.3 Condition ii): Frequency-Domain Robustness

The magnitudes of both $|G_c(j\omega)|$ and $1/|\Delta_G(j\omega)|$ were evaluated across the frequency range $\omega \in [10^{-3},\ 10^5]$ rad/s:

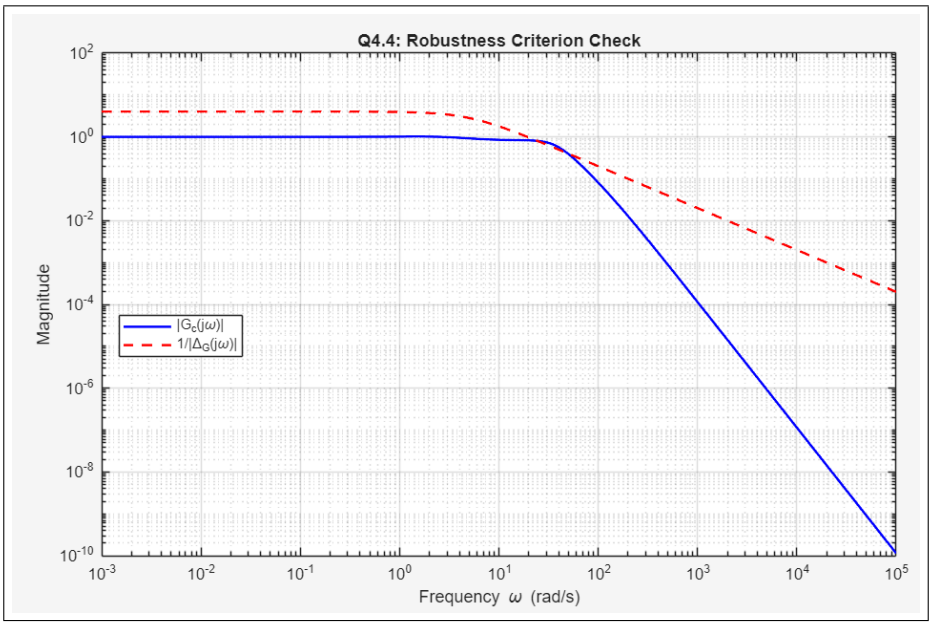

Figure 16: Robustness criterion check. The blue curve doesn't remain below the red dashed curve at all frequencies so is not robustly stable.

The plot reveals that the two curves intersect at approximately $\omega \approx 24.4$ rad/s. Past this frequency, the closed-loop magnitude $|G_c(j\omega)|$ exceeds the robustness bound $1/|\Delta_G(j\omega)|$. The worst-case violation occurs at $\omega \approx 33.4$ rad/s, where the margin between the bound and the closed-loop magnitude is -0.097. Evidently, this condition has not been passed.

```
Condition ii): |Gc(jw)| < 1/|Delta_G(jw)| for all w
The robustness criterion is not satisfied.
Violation occurs at 3812 frequency points.
First violation at w = 24.3693 rad/s
Worst violation: margin = -0.097030 at w = 33.4107 rad/s
```

Figure 17: Condition ii) result, violation frequency, and worst-case margin

4.4 Conclusion

Since condition ii) was not passed, robust stability under the given model $\Delta_G(s) = (s + 5)/20$ is not guaranteed. However, noting that this robustness criterion is only a sufficient condition, this does not necessarily prove that the system is unstable in totality, it only says that stability cannot be certified by this test. In reality, considering that the violation margin is relatively small, -0.097) [Figure: 17], so the system may in fact actually remain stable.

5 Assignment 4.5: State-Space & Pole Placement

5.1 State-Space Description

First, we need to find the state variables. From the assignment description and Figure 3, we can measure three things. Therefore, we set them as our state variables:

- $x_1 = \theta_L$ (Blade pitch angle)
- $x_2 = \dot{\theta}_m$ (Motor angular velocity)
- $x_3 = I_a$ (Armature current)

By looking at the block diagram, we can write the differential equations for the system:

$$\dot{x}_1 = n \cdot x_2 \tag{17}$$

$$\dot{x}_2 = -\frac{b}{J}x_2 + \frac{K_\tau}{J}x_3 \tag{18}$$

$$\dot{x}_3 = -\frac{K_m}{L_m}x_2 - \frac{R_m}{L_m}x_3 + \frac{1}{L_m}u \tag{19}$$

Then, we put them into the standard matrix form $\dot{\mathbf{x}} = A\mathbf{x} + Bu$ and $y = C\mathbf{x} + Du$:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & n & 0 \\ 0 & -\frac{b}{J} & \frac{K_\tau}{J} \\ 0 & -\frac{K_m}{L_m} & -\frac{R_m}{L_m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \frac{1}{L_m} \end{bmatrix} u \tag{20}$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad D = 0 \tag{21}$$

5.2 Controller Design

We need to design a full state feedback controller: $u = -K\mathbf{x} + k_r r$.

To meet the requirements, I used the following logic:

1. **Overshoot** $< 5\%$**:** I choose the damping ratio $\zeta = 0.7$, which can keep the overshoot safe.
2. **As fast as possible:** The system needs to be fast, but the control voltage $|u(t)|$ cannot exceed $u_{max} = 10326$ V.

To find the fastest speed, I used a `for` loop in MATLAB to test the natural frequency (ω_n) from slow to fast. The loop stops when the maximum voltage hits the 10326 V limit. The fastest natural frequency I found is $\omega_n = 33.00$ rad/s.

I put the two dominant poles at $p_{1,2} = -23.10 \pm 23.56j$. I also need a third pole, so I put it far away on the left at $p_3 = -115.50$, so it will not affect the system response too much.

Using the MATLAB `place` function, I calculated the feedback matrix K. After that, I calculated k_r to make the static gain equal to 1. The results are:

$$K = \begin{bmatrix} 9815.6293 & 7.9752 & 3.7646 \end{bmatrix} \tag{22}$$

$$k_r = 9815.6293 \tag{23}$$

5.3 Results and Plots

From the MATLAB simulation, the new rise time is very fast (0.0674 s). The overshoot is 4.36%, which is successfully less than 5%. Also, the static gain is exactly 1.

Figure 18 shows the control signal. The maximum voltage is 9815.63 V. It is very close to the limit, but it is safe and strictly below 10326 V. Figure 19 shows the step response of the closed-loop system.

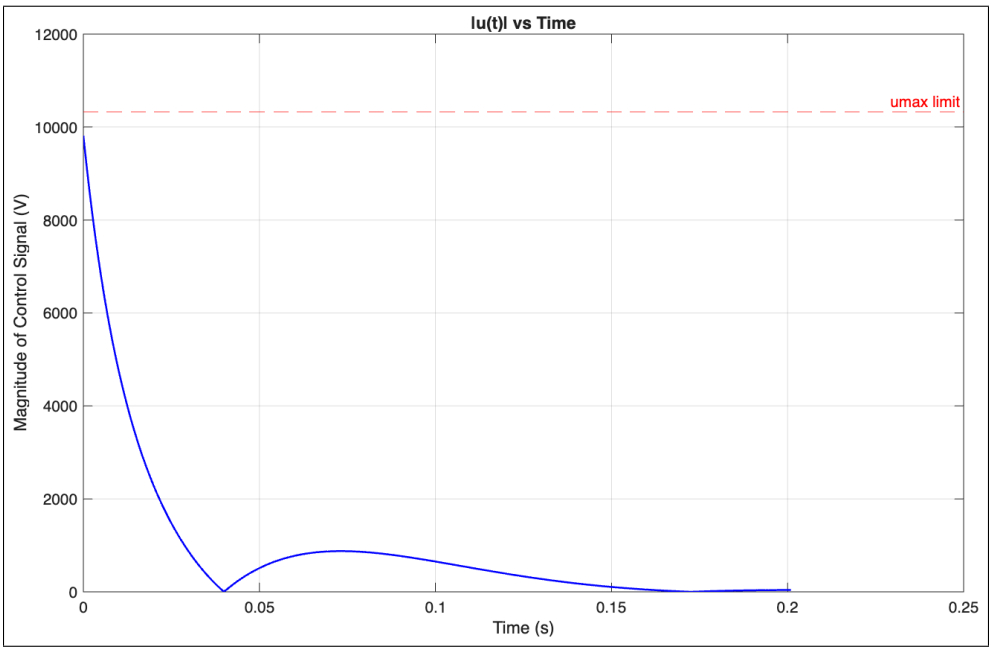

Figure 18: Control voltage $|u(t)|$ over time. It is below the u_{max} red line.

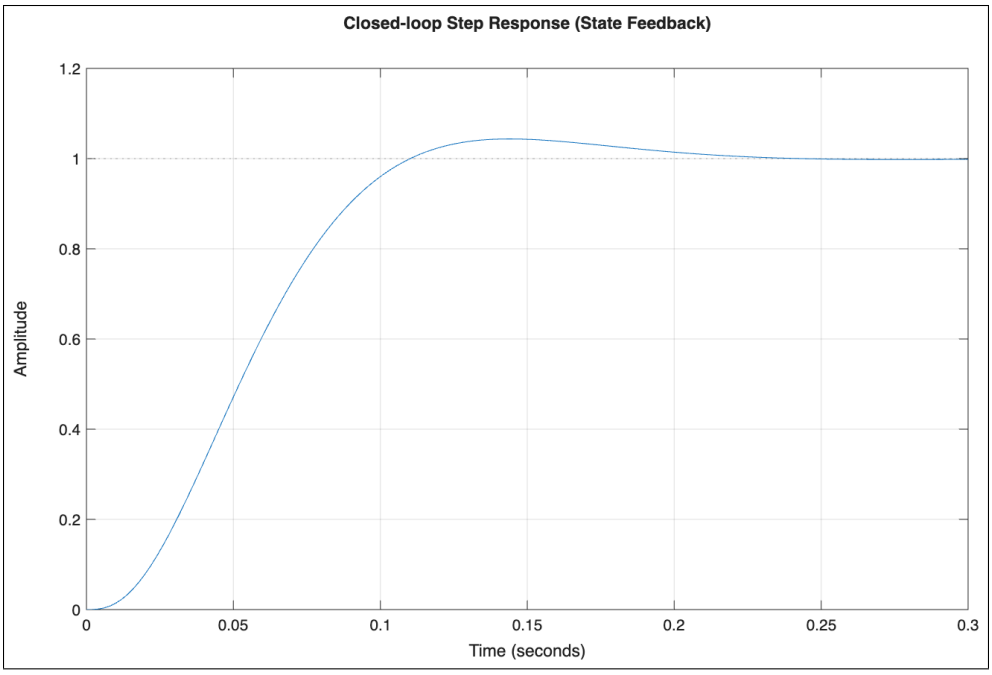

Figure 19: Step response of the new system.